

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 6 of 10

The Tds Equations and Entropy Change of Ideal Gases and Pure Substances

The two Tds equations · entropy change of an ideal gas · the standard processes · polytropic processes · solids, liquids and phase changes · property tables

Dr. Anantha Raman L., M.E., Ph.D. · Associate Professor, Department of Mechanical Engineering, Rajalakshmi Institute of Technology, Chennai

In this presentation

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2. The second Tds equation
3. Why they hold for any process
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3. Solids and liquids
4. Phase changes and property tables

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The Tds equations

The first Tds equation

$$\delta Q_{\text{rev}} = dU + p dV \quad \text{and} \quad \delta Q_{\text{rev}} = T dS$$

$$T dS = dU + p dV \quad \quad T ds = du + p dv$$

- Combine the first law for a reversible process with the definition of entropy

The second Tds equation

$$H = U + pV \Rightarrow dH = dU + p dV + V dp$$

$$T dS = dH - V dp \quad T ds = dh - v dp$$

- Substitute $dU + p dV = dH - V dp$ into the first equation

Valid for any process

Both Tds equations contain only properties (T, s, u, h, p, v), so they hold for any process – reversible or irreversible – between two equilibrium states.

- They are relations among properties, not statements about a path
- Working forms: $ds = du/T + (p/T) dv$ and $ds = dh/T - (v/T) dp$, with the equation of state giving p/T or v/T

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Entropy change of an ideal gas

Applying the Tds equations to an ideal gas

- $du = c_v dT$, $dh = c_p dT$, $p/T = R/v$, $v/T = R/p$
- First Tds equation: $ds = c_v dT/T + R dv/v$
- Second Tds equation: $ds = c_p dT/T - R dp/p$
- Integrate with constant specific heats

Entropy change of an ideal gas

$$\begin{aligned}
 s_2 - s_1 &= c_v \ln (T_2/T_1) + R \ln (v_2/v_1) \\
 &= c_p \ln (T_2/T_1) - R \ln (p_2/p_1) \\
 &= c_v \ln (p_2/p_1) + c_p \ln (v_2/v_1)
 \end{aligned}$$

- Third form: substitute $T_2/T_1 = p_2 v_2 / (p_1 v_1)$ and use $c_v + R = c_p$; multiply by m for $S_2 - S_1$

Entropy change in the standard processes

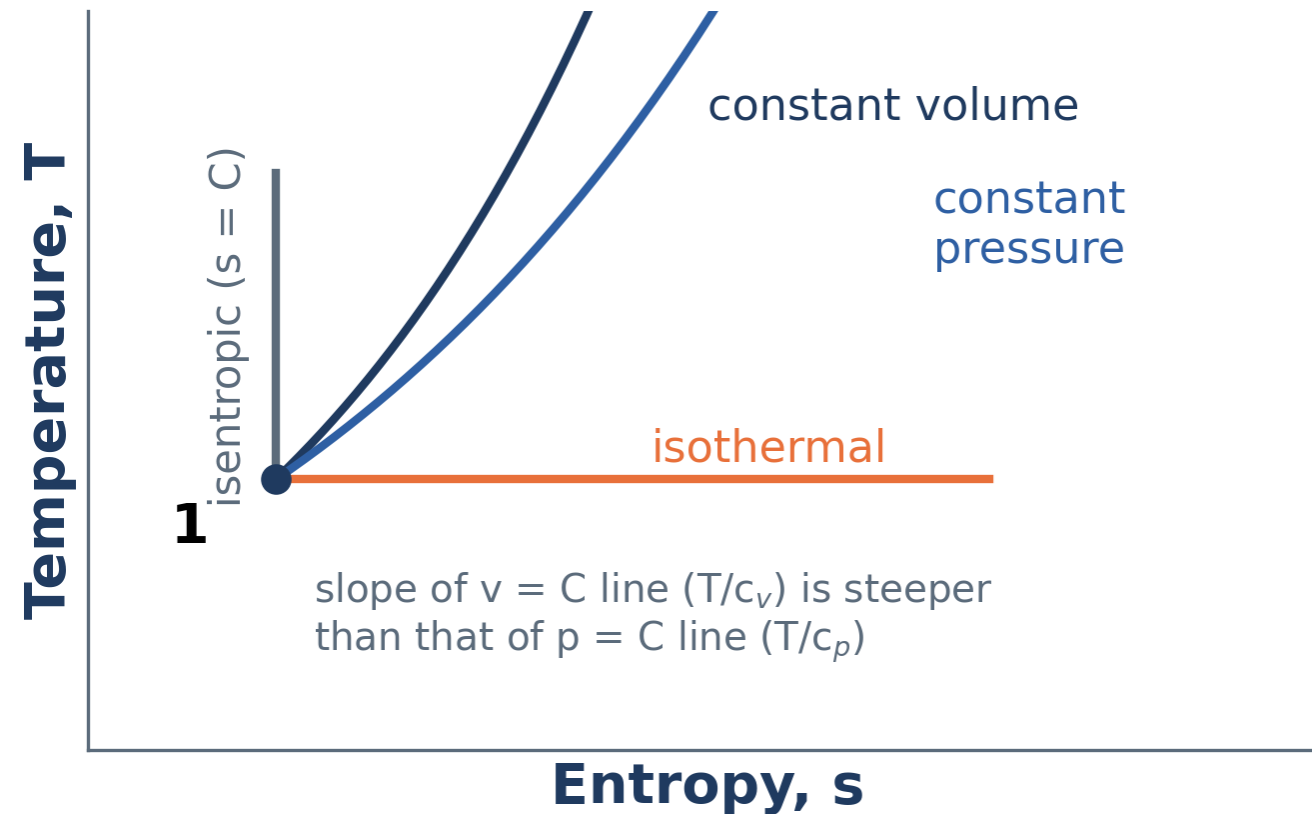
Process	Law	S2 – S1	Q1–2 (reversible)
Constant volume	$v = C$	$mc_v \ln (T_2/T_1) = mc_v \ln (p_2/p_1)$	$mc_v (T_2 - T_1)$
Constant pressure	$p = C$	$mc_p \ln (T_2/T_1) = mc_p \ln (V_2/V_1)$	$mc_p (T_2 - T_1)$
Isothermal	$T = C$	$mR \ln (V_2/V_1) = mR \ln (p_1/p_2) = Q/T$	$mRT \ln (V_2/V_1)$
Reversible adiabatic	$pV^\gamma = C$	0 (isentropic)	0
Polytropic	$pV^n = C$	$mc_v [(n - \gamma)/(n - 1)] \ln (T_2/T_1) = mc_n \ln (T_2/T_1)$	$mc_n (T_2 - T_1)$

The polytropic process

$$c_n = c_v (n - \gamma)/(n - 1) \quad S_2 - S_1 = mc_n \ln (T_2/T_1)$$

- From $s_2 - s_1 = c_v \ln (T_2/T_1) + R \ln (v_2/v_1)$ with $Tv^{n-1} = C$ and $R = c_v (\gamma - 1)$
- For $1 < n < \gamma$ (compression with cooling) c_n is **negative**: temperature rises while heat is rejected and entropy falls

Lines of constant v and p on the T – s plane



$T = T_1 \exp[(s - s_1)/c_v]$ and $T = T_1 \exp[(s - s_1)/c_p]$ – exponential curves; the constant-volume line is the steeper

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Solids, liquids and phase changes

Solids and liquids

$$S_2 - S_1 = mc \ln (T_2/T_1) \quad (\text{constant specific heat})$$

- Specific volume nearly constant, so the $p dv$ term vanishes and $c_p \approx c_v = c$
- $ds = c dT/T$

Phase change at constant pressure

$$S_2 - S_1 = m h_{fg} / T_{\text{sat}} \quad (\text{evaporation}) \quad S_2 - S_1 = m$$

$$h_{sf} / T_{\text{melt}} \quad (\text{melting})$$

- Temperature stays constant during melting or evaporation, so $\Delta S = \text{latent heat} \div \text{absolute temperature}$
- Vapour with $c_p = a + bT + cT^2$: $s_2 - s_1 = a \ln (T_2/T_1) + b (T_2 - T_1) + (c/2)(T_2^2 - T_1^2)$

Entropy of a pure substance from tables

- Wet mixture: $s = s_f + x s_{fg}$
- Compressed liquid: $s \approx s_f$ at the same temperature
- Superheated vapour: s tabulated against p and T
- Datum: $s = 0$ for saturated water at the triple point (Unit III)

Summary

- $T ds = du + p dv = dh - v dp$ – true for any process
- Ideal gas: $\Delta s = c_v \ln (T_2/T_1) + R \ln (v_2/v_1)$
- or $\Delta s = c_p \ln (T_2/T_1) - R \ln (p_2/p_1)$
- Isothermal Q/T ; isentropic 0; polytropic $mc_n \ln (T_2/T_1)$
- Solids and liquids $mc \ln (T_2/T_1)$; phase change $m h_{fg}/T_{sat}$
- Pure substances: s from the steam tables

Next: The principle of increase of entropy

- $dS \geq \delta Q/T$ and entropy generation
- $\Delta S_{\text{universe}} \geq 0$; entropy generated in common irreversible processes
- Applications of the second law
- Numerical problems: Worked Examples 2.7 – 2.10 in the lecture notes